

## KAIRAKUM, Tajikistan

WMO: 385980

Lat: 40.267N

Lon: 69.817E

Elev: 347

StdP: 97.23

Time Zone: 5.00 (E05)

Period: 10-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-----|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |     |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p) |
| 1             | -7.3       | -4.9 | -10.4                         | 1.6 | -6.8 | -7.6 | 2.1 | -4.0 | 15.6                  | 2.6  | 14.4 | 1.7  | 5.5                   | 270  | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.6                   | 36.2            | 22.0 | 34.8 | 21.3 | 33.6 | 21.0 | 24.4                | 32.9 | 23.6 | 32.1 | 22.7 | 31.1 | 3.7                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 21.7                            | 17.1 | 30.8 | 20.7 | 16.0 | 29.7 | 19.7 | 15.0 | 28.7 | 76.1          | 33.1 | 72.5 | 32.3 | 69.2 | 31.1 | 30.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 13.1          | 11.3 | 9.9 | DB                         | -9.3  | 39.6               | 4.2 | 1.4                                                | -12.3 | 40.6       | -14.8 | 41.4       | -17.2 | 42.2       | -20.2 | 43.1 | (4) |
| (5)               |      |     | WB                         | -10.1 | 27.1               | 4.0 | 1.6                                                | -13.0 | 28.3       | -15.3 | 29.3       | -17.6 | 30.2       | -20.5 | 31.4 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 15.1  | 2.1   | 3.0   | 9.9   | 16.2  | 22.3  | 26.5  | 28.0  | 26.0  | 21.6  | 14.5  | 7.1   | 2.8  | (6)  |
| (7)  |                                                                                  | DBStd       | 9.97  | 2.79  | 4.13  | 4.16  | 3.71  | 2.55  | 2.35  | 2.01  | 2.23  | 2.67  | 3.89  | 3.76  | 3.31 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 834   | 245   | 198   | 54    | 3     | 0     | 0     | 0     | 0     | 0     | 9     | 101   | 223  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 2228  | 503   | 431   | 261   | 81    | 2     | 0     | 0     | 0     | 5     | 127   | 338   | 481  | (9)  |
| (10) |                                                                                  | CDD10.0     | 2686  | 0     | 1     | 52    | 191   | 383   | 496   | 558   | 497   | 348   | 148   | 13    | 1    | (10) |
| (11) |                                                                                  | CDD18.3     | 1037  | 0     | 0     | 1     | 17    | 126   | 246   | 299   | 238   | 102   | 8     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 11084 | 0     | 0     | 1     | 113   | 905   | 2659  | 3811  | 2646  | 894   | 54    | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 4953  | 0     | 0     | 0     | 23    | 247   | 1164  | 1988  | 1247  | 279   | 6     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 4.0   | 5.0   | 4.8   | 4.4   | 3.6   | 3.9   | 3.7   | 3.3   | 2.9   | 3.1   | 4.0   | 4.5   | 5.1  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 162   | 16    | 18    | 26    | 29    | 20    | 5     | 3     | 1     | 2     | 15    | 14    | 17   | (15) |
| (16) |                                                                                  | PrecMax     | 330   | 58    | 49    | 74    | 85    | 68    | 24    | 22    | 10    | 23    | 90    | 60    | 62   | (16) |
| (17) |                                                                                  | PrecMin     | 77    | 0     | 2     | 2     | 1     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (17) |
| (18) |                                                                                  | PrecStd     | 57    | 12    | 11    | 18    | 20    | 16    | 6     | 5     | 3     | 5     | 19    | 13    | 14   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 9.2   | 14.9  | 21.5  | 29.6  | 33.2  | 37.4  | 39.5  | 36.7  | 33.2  | 27.5  | 19.3  | 12.0 | (19) |
| (20) |                                                                                  |             | MCWB  | 5.8   | 9.4   | 14.0  | 17.5  | 18.9  | 21.2  | 22.4  | 23.6  | 20.1  | 17.5  | 13.9  | 8.7  | (20) |
| (21) |                                                                                  | 2%          | DB    | 7.5   | 11.5  | 18.9  | 26.5  | 31.1  | 35.0  | 37.1  | 35.1  | 31.2  | 24.8  | 16.5  | 9.6  | (21) |
| (22) |                                                                                  |             | MCWB  | 4.8   | 8.0   | 13.2  | 16.3  | 18.8  | 20.0  | 22.1  | 22.2  | 19.6  | 16.6  | 12.3  | 6.6  | (22) |
| (23) |                                                                                  | 5%          | DB    | 6.4   | 9.6   | 17.2  | 24.1  | 29.3  | 33.5  | 35.4  | 33.9  | 29.7  | 22.7  | 14.5  | 8.2  | (23) |
| (24) |                                                                                  |             | MCWB  | 4.2   | 6.6   | 12.2  | 16.0  | 18.6  | 20.0  | 21.7  | 21.8  | 18.8  | 15.4  | 11.0  | 5.7  | (24) |
| (25) |                                                                                  | 10%         | DB    | 5.6   | 8.2   | 15.7  | 22.4  | 27.7  | 32.1  | 34.2  | 32.5  | 28.2  | 21.0  | 12.7  | 7.1  | (25) |
| (26) |                                                                                  |             | MCWB  | 3.6   | 5.7   | 11.6  | 15.3  | 18.5  | 20.4  | 21.3  | 21.3  | 17.9  | 14.6  | 9.4   | 5.0  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 6.7   | 10.1  | 14.9  | 18.8  | 21.8  | 24.6  | 25.7  | 25.4  | 22.1  | 18.7  | 14.3  | 8.8  | (27) |
| (28) |                                                                                  |             | MCDWB | 8.5   | 13.9  | 19.9  | 25.7  | 28.0  | 31.8  | 33.4  | 33.9  | 30.4  | 25.4  | 18.4  | 11.7 | (28) |
| (29) |                                                                                  | 2%          | WB    | 5.2   | 8.4   | 13.6  | 17.4  | 20.8  | 23.5  | 24.8  | 23.9  | 20.7  | 17.4  | 12.7  | 7.1  | (29) |
| (30) |                                                                                  |             | MCDWB | 6.7   | 11.2  | 17.7  | 23.9  | 27.8  | 30.8  | 33.4  | 32.6  | 29.1  | 23.4  | 16.2  | 9.1  | (30) |
| (31) |                                                                                  | 5%          | WB    | 4.5   | 6.9   | 12.7  | 16.6  | 20.1  | 22.7  | 23.9  | 23.0  | 19.8  | 16.2  | 11.3  | 6.0  | (31) |
| (32) |                                                                                  |             | MCDWB | 5.9   | 9.1   | 16.1  | 22.0  | 26.6  | 30.1  | 32.9  | 32.1  | 28.0  | 21.6  | 14.2  | 7.7  | (32) |
| (33) |                                                                                  | 10%         | WB    | 3.8   | 5.8   | 11.9  | 15.9  | 19.5  | 21.9  | 23.1  | 22.1  | 18.8  | 15.2  | 9.8   | 5.1  | (33) |
| (34) |                                                                                  |             | MCDWB | 5.3   | 7.9   | 15.3  | 21.3  | 25.9  | 29.3  | 31.8  | 30.7  | 26.6  | 19.9  | 12.4  | 6.9  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | MDBR        | 4.3   | 5.6   | 7.4   | 9.6   | 10.4  | 11.0  | 12.6  | 13.0  | 13.3  | 10.2  | 7.5   | 4.9   | (35) |      |
| (36) |                                                                                  | 5% DB       | MCDBR | 5.1   | 7.3   | 9.6   | 12.8  | 13.1  | 13.7  | 14.9  | 15.1  | 15.1  | 13.0  | 9.6   | 5.8  | (36) |
| (37) |                                                                                  |             | MCWBR | 3.5   | 4.6   | 4.5   | 5.2   | 4.9   | 5.2   | 5.8   | 6.2   | 6.5   | 6.3   | 6.0   | 3.8  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 4.7   | 6.9   | 8.4   | 10.7  | 10.8  | 11.0  | 13.3  | 14.5  | 15.1  | 11.8  | 9.0   | 5.6  | (38) |
| (39) |                                                                                  |             | MCWBR | 3.5   | 4.7   | 4.4   | 4.8   | 5.0   | 6.1   | 6.7   | 6.8   | 7.2   | 6.3   | 5.8   | 4.0  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.364 | 0.414 | 0.479 | 0.498 | 0.492 | 0.491 | 0.476 | 0.452 | 0.439 | 0.478 | 0.437 | 0.369 | (40) |      |
| (41) |                                                                                  | taud        | 2.157 | 2.037 | 1.924 | 1.931 | 2.001 | 2.010 | 2.053 | 2.114 | 2.082 | 1.921 | 1.987 | 2.130 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 762   | 772   | 763   | 782   | 797   | 797   | 804   | 814   | 793   | 691   | 660   | 717   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 106   | 138   | 173   | 183   | 175   | 173   | 164   | 151   | 146   | 153   | 125   | 101   | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.96  | 2.76  | 3.92  | 5.18  | 6.43  | 7.42  | 7.51  | 6.83  | 5.58  | 3.64  | 2.25  | 1.70  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.19  | 0.34  | 0.34  | 0.55  | 0.40  | 0.31  | 0.26  | 0.31  | 0.18  | 0.35  | 0.28  | 0.21  | (45) |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (11 neighbors) |  | N/A   | N/A     | N/A    | +0.50   | N/A   | N/A   | N/A         | N/A     | +86     | +63     |

Nomenclature: See separate page